

DDoS Anomaly Detection in Network Traffic Using Quantum-Inspired Machine Learning

QCentroid Challenge - Co-developed with GSMA

The challenge

Identifying anomalous traffic behaviour is a key challenge in modern networks. Operators face increasing exposure to cyber threats alongside rapidly growing data volumes and highly dynamic user demand. Malicious actors can degrade service quality, exhaust network resources, and in some cases compromise sensitive infrastructure if not promptly identified.

In this challenge, you are given a series of datasets representing network activity in a campus network environment. **Your task is to detect Distributed Denial of Service (DDoS) attacks.**

Your task

Each dataset family contains 12 benign datasets and 12 attack datasets — the dataset-level labels are provided. A single flow row is not enough to determine whether traffic is malicious. The signal lives at the **time-window** level: short intervals (e.g. one minute) within an attack dataset where flow-level patterns deviate from the benign baseline.

Within the attack datasets:

1. **Identify the time windows that contain DDoS activity.** A window is a slice of consecutive flows over a short temporal interval; your job is to flag which windows contain a burst.
2. **Identify the malicious source IPs** associated with the activity in those windows.

Your solution should use **quantum-inspired machine learning**. The quantum component does not need to cover the entire pipeline – you may apply it to a specific sub-task where it is genuinely well suited, and use classical methods elsewhere. What matters is that you can justify why you chose a quantum or quantum-inspired technique for the part of the problem you applied it to.

Background: DDoS in flow-level data

In network monitoring, packets are aggregated into **flows** that share source IP, destination IP, source port, destination port, and network protocol. Each row in the datasets represents one flow.

A DDoS attack overwhelms a target with a large number of requests originating from many sources. A single flow does not carry enough information to call an attack on its own — the signature is collective, visible only when you look at a population of flows within a short time window. Within an attack window, the typical pattern combines:

- High flow count
- High total packets
- Many source IPs
- One or few destination IPs
- One or few destination ports
- Short flow durations
- Small or moderate packet sizes
- High packets / bytes per second

This is provided as background only. How you exploit (or ignore) these signals is your choice.

The data

You are given **two dataset families**, evaluated separately:

- **Family A:** benign and attack traffic from the same source dataset (NF-UNSW-NB15-v3). Attack flows are DoS rows recast as DDoS, with source IPs replaced from a synthetic bot pool of 50–100 addresses.
- **Family B:** benign traffic from NF-UNSW-NB15-v3; attack traffic from native DDoS rows in NF-CSE-CIC-IDS2018.

Each family contains 24 datasets:

Split	Normal	Attack
Training	8	8
Validation	2	2
Test	2	2

Each dataset represents a synthetic 15-minute window of activity with **100,000 flow rows**. Attack datasets contain **2–4 DDoS bursts** (ramp-up, peak, and taper phases) seeded into otherwise-normal traffic. Seeded attack rows account for 1.5–2.5% of each attack dataset (approximately 1,500–2,500 rows out of 100,000).

Schema (per flow)

- **src_ip, dst_ip** — source and destination IP
- **src_port, dst_port** — source and destination port

- `protocol` — TCP, UDP, or ICMP
- `duration` — flow duration in seconds
- `packets_per_second`, `bytes_per_second` — flow rates
- `inter_packet_arrival_mean`, `inter_packet_arrival_std` — packet timing
- `total_packets`, `total_bytes` — flow volumes
- `packet_size_avg`, `packet_size_std` — packet size distribution
- `outbound_byte_ratio` — share of bytes flowing outbound

Audit fields (`scenario`, `split`, `dataset_id`, `row_in_window`, `is_seeded_ddos`, `burst_id`, `burst_phase`, `source_dataset`) are included for inspection and validation. **They must not be used as model input features.**

Evaluation

Evaluation focuses on **the type of solution proposed and its technical justification**, not on raw numeric performance alone. A well-reasoned, well-implemented approach can outscore a higher-accuracy submission with a weak rationale. The judging panel will assess:

- **Appropriateness of the quantum technique** for the chosen task. The quantum component does not need to address the whole problem — it should be applied where it makes sense.
- **Justification of the choice.** Why this technique, for this sub-task, in preference to alternatives.
- **Realism under simulator and hardware constraints.** Whether the chosen approach could plausibly offer an advantage given the resources available during the event, and whether the team is honest about its limitations.
- **Technical quality of the implementation.** Clean code, sound experimental setup, and rigorous validation are a plus.

Time-window detection accuracy and source-IP identification are reported as supporting evidence, not as the primary criterion.

What you receive

Participants who select this challenge will be provided with:

- The **full dataset suite** (training, validation, and test) for both families.
- **Instructions** for working with the data.
- A **starter notebook** containing the technical information required to begin (data loading, schema details, submission format).

Running on the **QCentroid platform** is optional. You may use the platform's simulators and quantum backend if it suits your approach, or work in your own environment. Full technical details and dataset access will be shared with participants once the challenge is selected.

Submission

By the end of the event window, submit:

- Your **code**.
- A **short writeup describing your approach**, the sub-task you applied a quantum technique to, why you chose it, and what you observed.
- A **short live presentation** or demo, depending on final event timing.